Galileo Engineering

CVP5-C -M12

Electronic Hydraulic Driver with CAN capability



Technical Specifications

Power Supply Voltage (abs. Min-max)	12-30Vdc
Rectified and filtered ripple voltage (max)	10%
Output Current (adjustable) (min-max)	0 – 2.5A
Current absorption (max) without load	30mA
Offset current (adjustable) (min-max)	0 – 0.6A
Current Regulation (CAN BUS)	± 1 mA (16 bit)
Maximum absorbed power (average)	35W
PWM Frequency (fixed)	5 kHz
Adjustable Ramp down slope	0.01 – 50 A/s
Adjustable Ramp up slope	0.01 – 50 A/s
Dither Frequency (fixed)	125 Hz
Dither Amplitude	0 to 20% of output current
Current stability (on temp. range)	± 5 mA
Operating temperature range	-20 ... +70 °C
Protection class CVP5-C M12	IP67
Weight	100g

Package	What	Where
Bag	PE-LD	Plastic

General Description

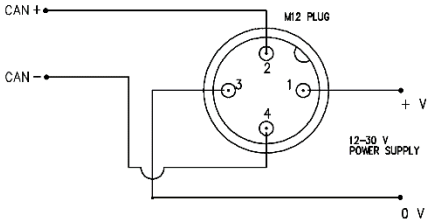
The CVP5-C-M12 Proportional Valve Controller is a compact and efficient electronic regulator designed for precision control of proportional valve solenoids. This controller seamlessly integrates with the CANopen® communication protocol and the Zephyr RTOS for robust performance and control, catering to a wide range of applications.
CVP5-C-M12 comes in a small and compact form factor with a M12 connector and is rated IP67 making it a versatile solution for both mobile and industrial hydraulic equipment.

Key Features:

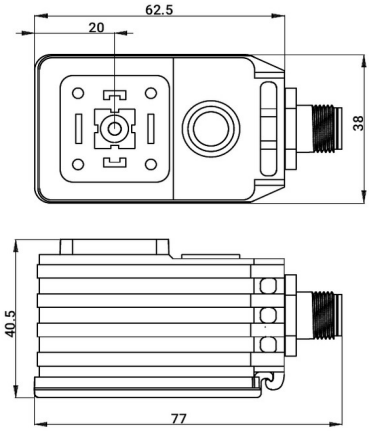
- **Robust Design:** Built to withstand harsh environments, the CVP5-C-M12 features protection against power supply polarity inversion and solenoid short circuits (5 ms detection time), ensuring longevity and reliability in demanding conditions.
- **Compact Form Factor:** The compact design enables easy installation and integration into a wide range of applications without consuming excessive space. Equipped with an M12 connector, the CVP5-C-M12 ensures easy integration into your system and holds an IP67 rating for protection against dust and water, making it versatile in various environments.
- **CANopen® Integration**: Seamlessly communicate with up other devices in your network using the CANopen® standard communication protocol, ensuring compatibility and interoperability within your system. The output current can be configured to be controlled by any other CANopen® device on the network, from the throttle of a joystick to a press of a button.
- **Fully Customizable:** Tailor the controller's settings to your specific needs, allowing for fine adjustments that optimize valve performance and system efficiency. Through CANopen® interface, you are allowed to configure MAX current, OFFSET current, Current Ramps and Dither Effect.
- **System Diagnostic:** Through the CAN interface it is possible to monitor real-time the output current and diagnostic state between open-circuit, short-circuit, MAX current reached and supply voltage limit.
- **Status Indication:** A three-color RGB LED indicator provides at-a-glance information about the system's status, simplifying monitoring and troubleshooting.
- **RTOS Powered:** Leveraging the Zephyr RTOS (Real-Time Operating System) provides real-time capabilities and efficient multitasking, enhancing the controller's performance and reliability.
- **Advanced Control:** Employing advanced PID logic, this controller offers precise and reliable valve control, meeting the demands of applications requiring precision and responsiveness.

Electrical Connections

PIN 1 BROWN +V POWER SUPPLY
PIN 2 WHITE CAN+
PIN 3 BLUE 0V POWER SUPPLY (GROUND)
PIN 4 BLACK CAN–



Dimension



Notes

The GND pin is otherwise marked as pin 4 inside the controller.

The power supply voltage must be in the 12 to 30 V dc range. It is necessary to power the system with rectified and filtered voltage. The use of a 4700 uF 35V electrolytic capacitor is recommended in order to filter the power voltage supply.

The electronic controller can drive valves with coil powered at 12 or 24 Vdc. To assure the nominal maximum current value of the coil the voltage supply of the controller must exceed the nominal voltage supply of the coil valve at least of 1,5V.

The controller is factory-calibrated to suit a wide spectrum of solenoid valves, making it versatile for various applications. However, certain specialized applications may necessitate fine-tuning of the PID control gains and voltage-current mapping to precisely align with the characteristics of the solenoid coil. This feature empowers users to achieve optimal performance tailored to their specific requirements, balancing both plug-and-play usability and the capacity for intricate customization.